GROUP 7

Statistical Modelling of the Ultimate Superhero Tournament

A data-driven clash of universes using Principal Component Analysis, K-Means Clustering, Logistic Regression, and Markov Chain transitions.

Submitted to: Prof. Jimut Bahan Chakraborty

Group Members:

- Abdul Arham Shamim
- Anjali
- Chahat Kamalni
- Jeptha Lalawmpuia
- Nandini Talwar



Problem Statement: Can Statistics Predict the Ultimate Superhero?

The Subjective Barrier

Superhero battles are traditionally decided by subjective fan opinions and favorite-character bias. However, superhero profiles have multiple measurable attributes like Strength, Speed, and Intelligence, enabling statistical analysis.

The Project Objective

To develop a data-driven superhero tournament model that uses statistical techniques to predict the outcomes of hypothetical 1v1 matchups. Pairwise battles are modeled using the overall score difference as a proxy winner target.

Methodology & Simulation

Using Logistic Regression to estimate win probabilities and identify which attributes matter most. The model then propagates progression odds through a stochastic tournament and simulates what-if entry barriers for new heroes.

CORE STRATEGIC QUESTION

Can statistical modelling turn superhero battles from subjective debates into a data-driven tournament, and what characteristics ultimately give a superhero the greatest probability of winning?

Beyond Fan Debates: Bringing Rigor to the Multiverse

THE SUBJECTIVE BARRIER

Recency Bias & Favoritism

- Arguments are heavily prone to recency bias (e.g., who starred in the latest blockbuster film).
- Subjective fan favoritism dominates, focusing on emotional attachment rather than analytical consistency.
- Character rankings ignore secondary uncertainty introduced by a tournament structure, where a hero's success depends on the exact seeding and match pathways.

THE ANALYTICAL SOLUTION

Rigor & Probability Models

- Standardized ratings in 6 categories: Intelligence, Strength, Speed, Durability, Power, and Combat.
- Applying statistical dimensionality reduction to handle attribute correlations cleanly.
- Propagating probabilities sequentially across a 16-player bracket to calculate realistic tournament outcomes, proving that 'strongest' and 'most likely champion' are distinct.

Filtering and Restructuring the Superhero Multiverse

RAW DATABASE SCALE

1,471

Bio-dataset rows containing biographical details, publisher indices, and multiple raw power categories across creators.

CLEANED CASSETTES

958

Unique character profiles retained after filtering exclusively to Marvel and DC, removing duplicates, and purging incomplete records.

PREDICTORS RETAINED

6

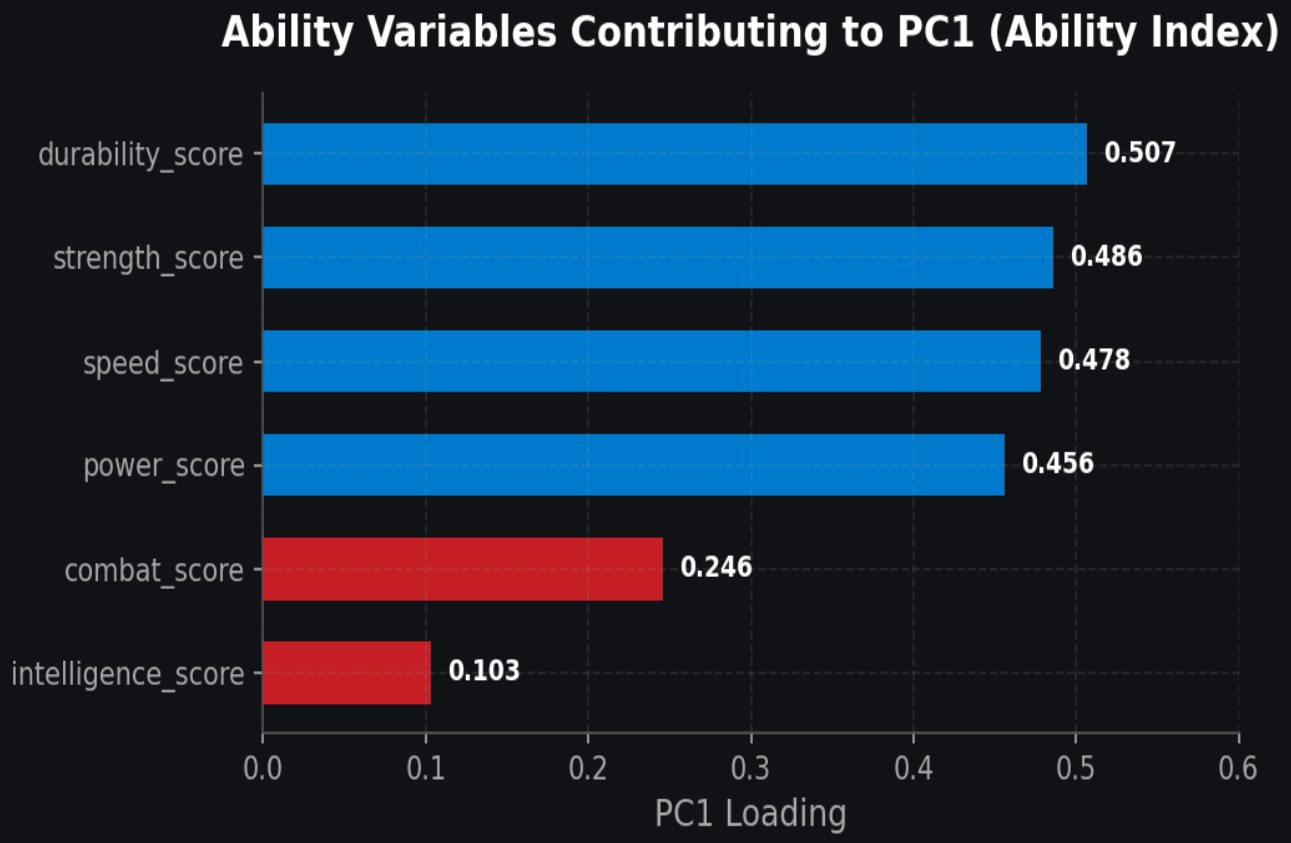
Standardized abilities: Intelligence, Strength, Speed, Durability, Power, and Combat used to model character-level capabilities.

PCA Compresses 6 Core Stats to Build a Single Ability Index

DIMENSIONALITY REDUCTION

Principal Component 1 Captures 50.4% Variance

- PCA evaluates if six highly correlated stats can be reduced to a smaller set of orthogonal components.
- PC1 captures 50.4% of total standardized variance, serving as our composite Ability Index.
- Standardization of all six abilities is crucial prior to projection to avoid scaling imbalances.
- Loading strengths show that Durability (0.507), Strength (0.486), and Speed (0.478) dominate PC1. Combat (0.246) and Intelligence (0.103) contribute weakly.

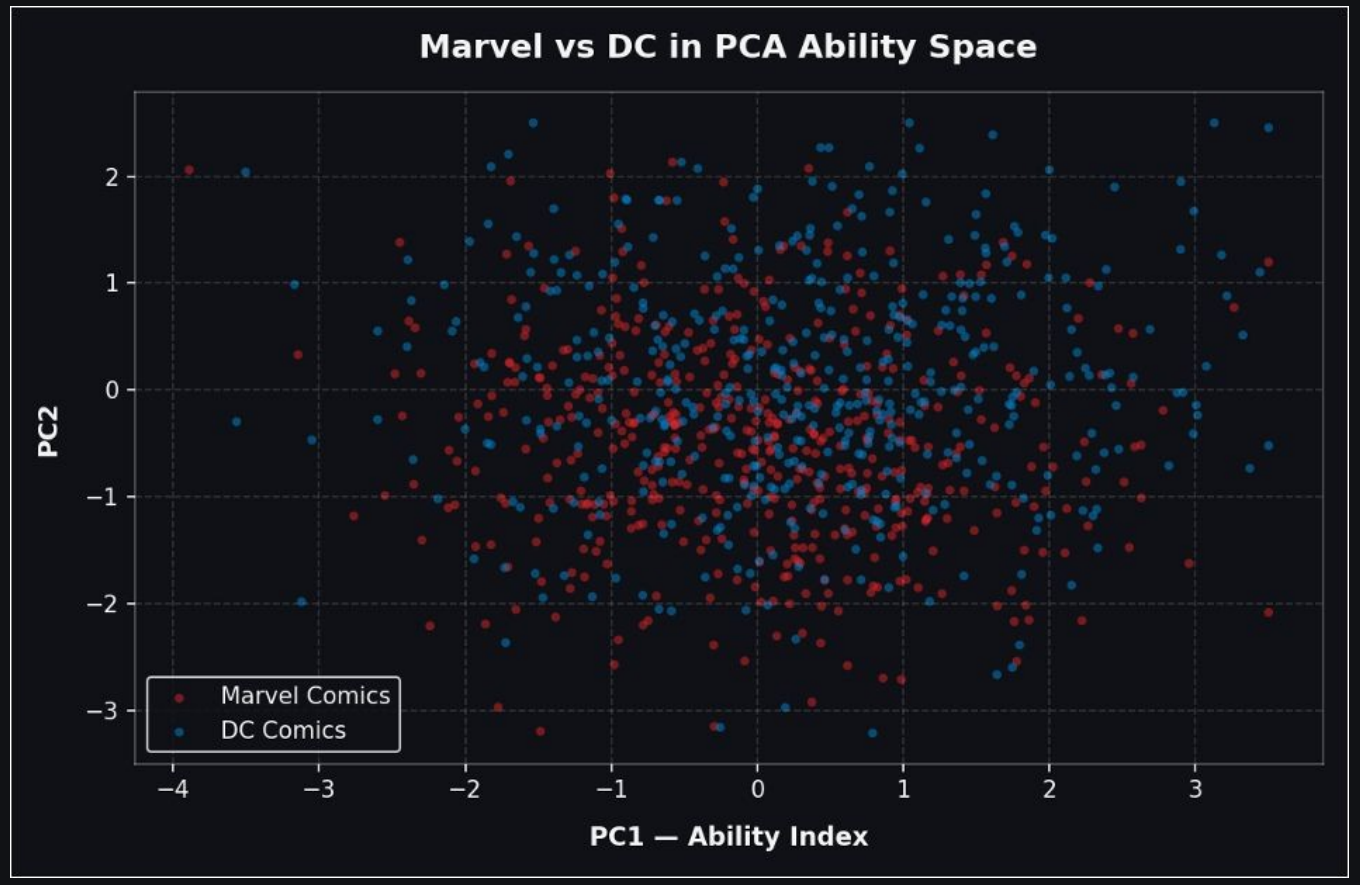


PCA Coordinate Space: Substantial Overlap Between Universes

COORDINATE SPACE VISUALIZATION

PC1 vs. PC2 Character Coordinate Space

- Plotting characters in the PC1-PC2 coordinate space allows us to visualize character patterns.
- PC1 represents raw physical/energy capacity (Ability Index), while PC2 explains 18.9% of variance, capturing intelligence/skill trade-offs.
- Substantial spatial overlap occurs between Marvel (red) and DC (blue), showing neither publisher holds an exclusive monopoly on extreme characters.
- Outliers show cosmic beings at the far right, and grounded characters at the lower-left.

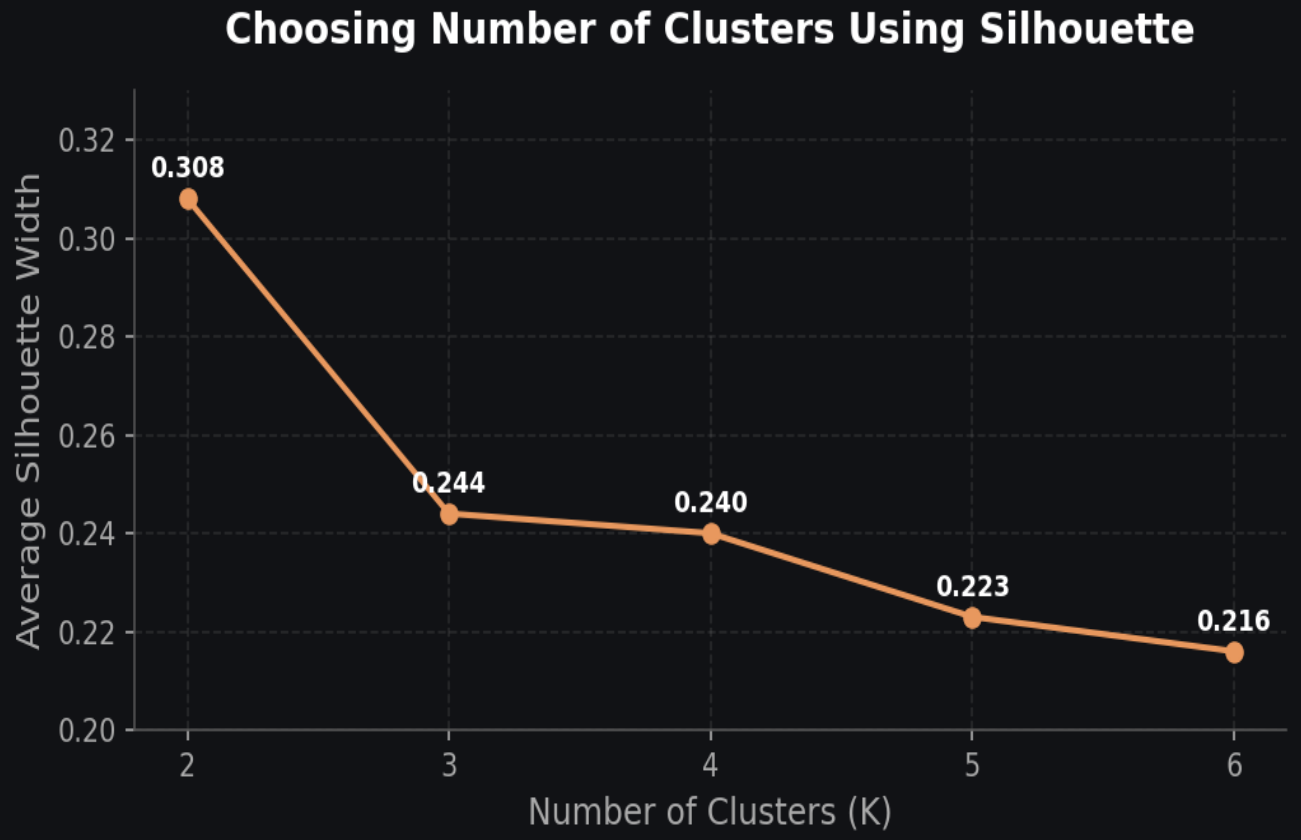


K-Means Partitions Characters Into 3 Combat Archetypes

EXPLORATORY PROFILING

Clustering Reveals 3 Character Archetypes

- K-Means is used alongside PCA to find natural, descriptive groupings of power profiles.
- Silhouette metric peaks at K=2 (~0.31), indicating two massive classes (grounded vs powerhouse). We select K=3 (~0.24) to isolate a richer mid-tier.
- Crucially, clustering is kept purely exploratory — it is not used to predict matchups.
- Cluster 1: Cosmic Powerhouses | Cluster 2: Street-Level / Skill-Based | Cluster 3: Mid-Tier / Specialists.

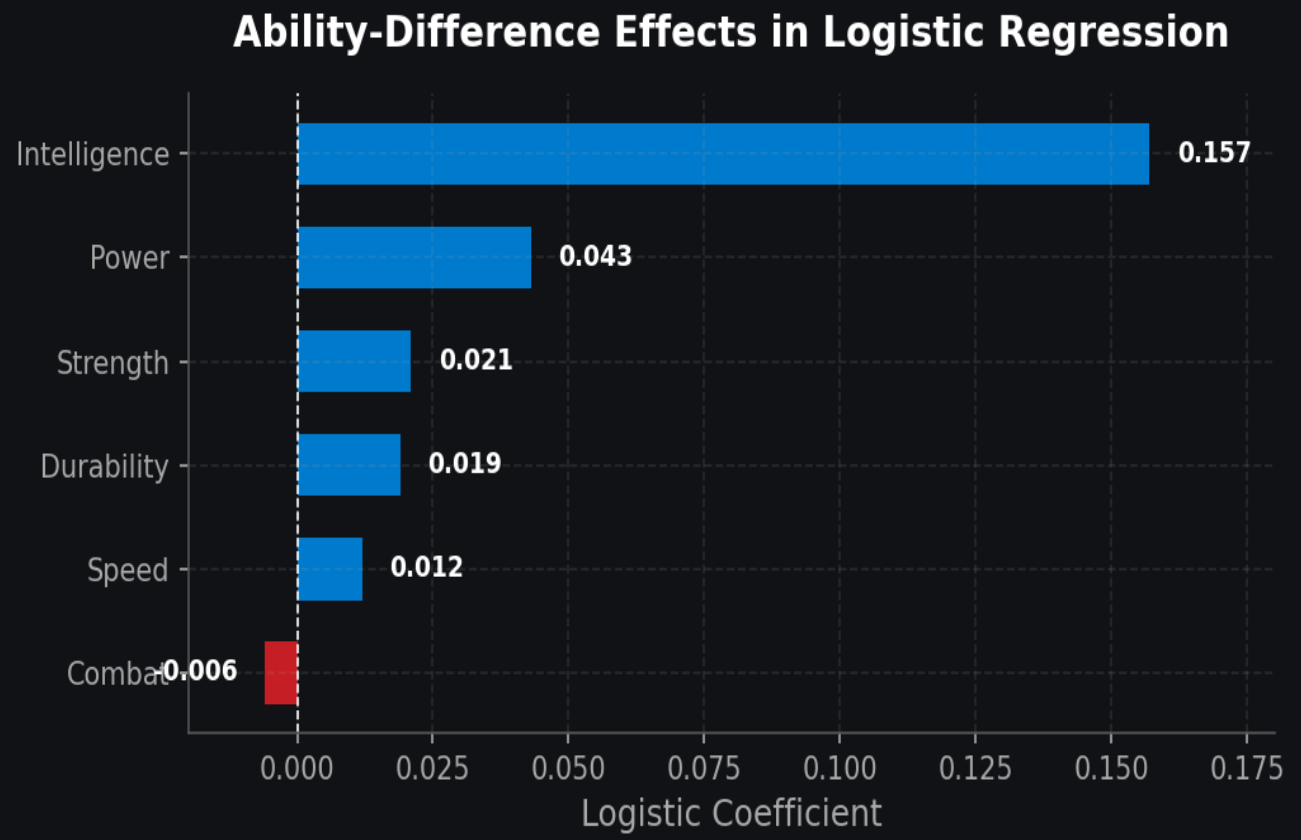


Logistic Model Reveals Intelligence Outweighs Combat Skill

PAIRWISE PROBABILITY MODEL

The Combat Paradox & Coefficients

- Pairwise training cases are constructed using difference in ability scores, using overall score delta as a proxy winner target.
- We fit a binomial generalized linear model (GLM) to convert linear predictors into bounded probabilities.
- Intelligence has an overwhelming positive effect (0.157 coefficient). Power (0.043) and Strength (0.021) follow.
- The Combat Paradox: Combat has a slight negative coefficient (-0.006), suggesting minor combat rating increases do not predict wins once physical power and brains are accounted for.



Markov Transitions Propagate Odds Across Bracket Stages

Sequential State Transitions

The tournament is modeled as a sequence of discrete transition steps: Round of 16 → Quarterfinals → Semifinals → Finals → Champion. A stochastic state vector represents the probability of a hero occupying the active position, which we aggregate across possible opponent pairings.

The Probability Axiom (QA Rule)

The initial pipeline computed independent pairwise matchups, which violated probability axioms (summing to >1.00). We enforce P(B wins) = 1 - P(A wins) to guarantee a strict probability-conserving sum of exactly 1.00 (100%) across all players.

SEQUENTIAL TOURNAMENT DECAY FLOW

Round of 16 ⟶ Quarterfinals ⟶ Semifinals ⟶ Finals ⟶ Champion

- Round of 16: 16 entrants sorted into fixed seed positions (8 Marvel vs 8 DC) based on PC1 score.
- Quarterfinal: Survival probability is determined by initial 1v1 win likelihoods.
- Semifinal & Finals: Paths are aggregated over all possible potential opponents carrying their respective survival weights.

Thor Claims the Statistical Crown with 13.8% Odds

STOCHASTIC CHAMPION

Why the Strongest Doesn't Always Win

- Cosmic King Thor is the ultimate modeled champion, securing a 13.8% championship probability.
- Rank vs. Odds Divergence: Thor is NOT the highest-ability character (Hercules, One-Above-All exceed him), yet has higher final odds.
- The Seeding Advantage: Thor's bracket seed grants him an easier initial branch, avoiding early-round roadblocks.
- Analytical Insight: This proves that raw ability ranking and final tournament odds are distinct.

Character	Creator	Championship Odds
Cosmic King Thor	Marvel Comics	13.8%
Black Adam	DC Comics	11.1%
One-Above-All	Marvel Comics	10.5%
All-father Hercules	Marvel Comics	10.3%

Tracing Survival Decay across Knockout Tournament Stages

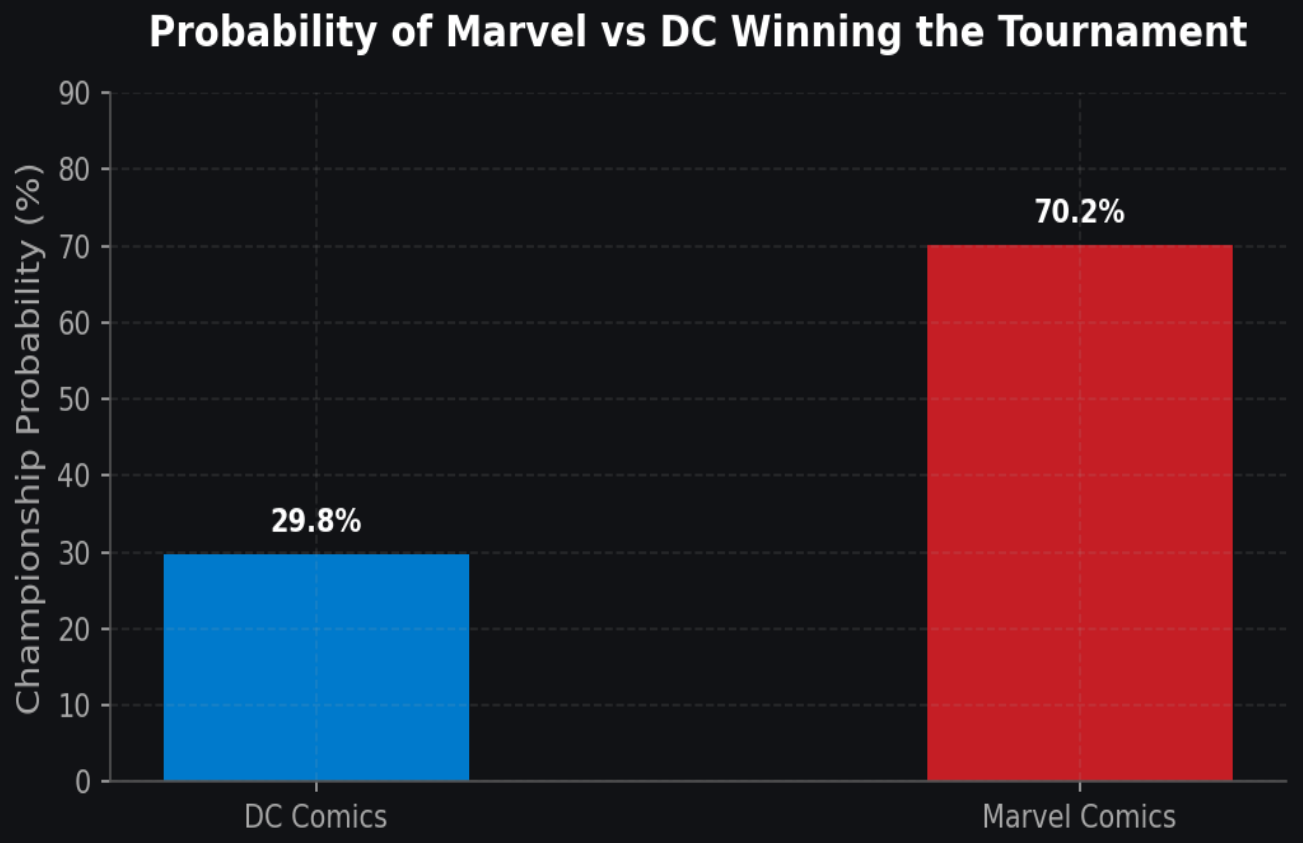
Top 8 Contenders	Quarterfinals	Semifinals	Finals	Tournament Champion
Cosmic King Thor (Marvel)	91%	50%	27%	13.8%
Black Adam (DC)	70%	38%	21%	11.1%
One-Above-All (Marvel)	70%	40%	20%	10.5%
All-father Hercules (Marvel)	70%	39%	20%	10.3%
Ragnarok (Marvel)	70%	37%	19%	9.7%
Dark Phoenix (Venomized) (Marvel)	52%	31%	18%	9.6%
Ultron (Final Form) (Marvel)	67%	31%	17%	8.5%
Great Evil Beast (DC)	55%	31%	16%	8.0%

Marvel Outpaces DC with 70.2% Championship Probability

AGGREGATED BRACKET POWER

Publishers Clash in the Tournament

- By summing character-level championship probabilities within each publisher, we resolve the classic debate.
- Marvel Comics secures 70.2% of aggregate championship probability, while DC holds 29.8%.
- No Canonical Claim: This is an artifact of bracket composition and seeding, not a claim that Marvel characters are canonically stronger.
- Distributed Strength: Marvel's powerful characters are distributed across seeds in a way that aggregates massive survival paths.

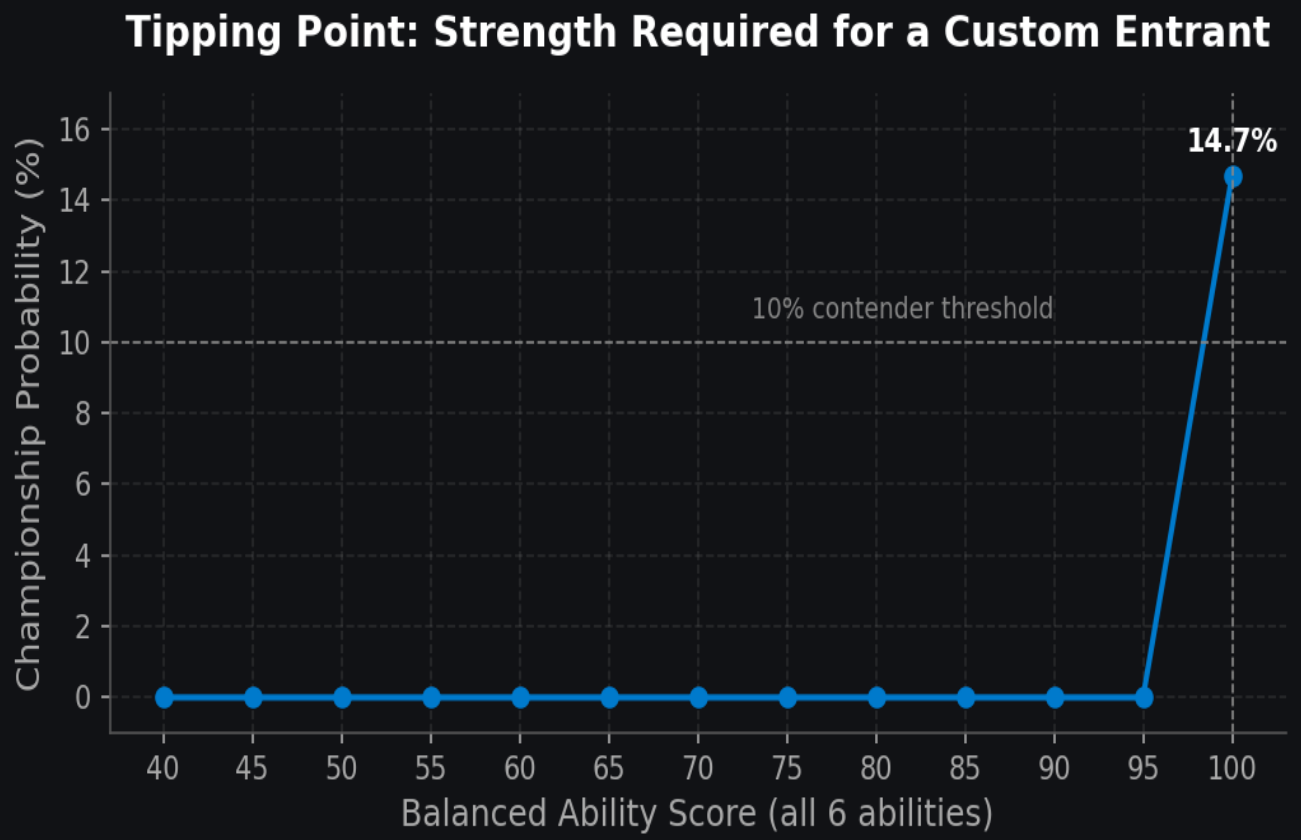


Balanced Entrants Must Hit Perfect 100/100 to Compete

SENSITIVITY ANALYSIS

The Tipping Point for New Characters

- We test model sensitivity by inserting a perfectly balanced new hero with identical ratings (40 to 100) across all six abilities.
- The new hero is projected into PCA space, sorted, and appended to the tournament (bumping the 16th player).
- The Tipping Point: Below 100/100, the entrant has exactly 0% championship probability because they fail to make the top-16 roster cut.
- At perfect 100/100 ratings, they instantly become the top threat with 14.7% champion odds.



Empirical Battle Data and Seeding Tests Define the Roadmap

EMPIRICAL BATTLE DATA

Moving Beyond Proxy Targets

Currently, no actual battle history exists in our database. Wins are trained using overall score differences as proxy targets. Collecting real victory logs from comic books will significantly improve predictor coefficients.

MONTE CARLO SEEDING

Isolating Seeding Luck

Seeding and brackets are currently fixed, leading to seeding bias (which benefited Thor). Run Monte Carlo simulations across randomized bracket shuffles to separate raw character power from pure placement luck.

PROBABILITY INTEGRITY

Retaining Rigorous Axioms

It is critical to preserve probability-consistency rules. Ensuring P(B wins) = 1 - P(A wins) maintains mathematical integrity, preventing sums from exceeding 100% in sequential transitions.

THE END

Thank You!